

Operator UI / TUI + twin renderer
UI → API writes; renderer displays state

REST / WebSocket + A2A
shared authenticated entry points

MCP interface
read-oriented subset; not all host tools

PLATFORM PROCESS

Agent orchestrator

harness adapters · team channels
lead / workers · tasks · memory
host tools + scoped dispatch
shared node / team / run IDs

intent

Binding + approval bridge

node semantics · read/write scope
manual / auto policy · approvals
agent-write admission + readback
records · separate backstop

write

Industrial supervisor

DAQ runtimes + gateway sweep
DCW dispatch + protocol drivers
alarms · line / recipe / run state
device-twin state publication

External CLI engines

separate processes; containment external

Separate persistent stores

platform DB · time series · objects · audit

Logical IDs; not atomic storage

Plant / protocol boundary

physical devices or seeded scenarios

Solid: request / control · Dashed: observation / record flow · PLC safety and hard real-time control remain external.